OPTIMAL EXPLICIT-FORMULA BOUNDS FOR ANALYTIC RANKS: THE LIMITING RADIX KERNEL

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Abstract. For an elliptic curve $E/\mathbb{Q}$ and a window $[-L, L]$, the radix kernel of Hackfort (2026) is the test function that is optimal for each curve, within a finite-dimensional space, for bounding the analytic rank of E under GRH by the explicit formula. Assuming GRH, we prove that the optimum over all admissible test functions with this support exists, is unique, equals the reciprocal of the value at the origin of the reproducing kernel of the Hilbert space defined by the Weil quadratic form, and is the limit of the finite-dimensional radix bounds, with an exact error identity. By Akhiezer's factorisation theorem, no continuous test function g with $\operatorname{supp} g \subset [-2L, 2L]$, $\hat{g} \geq 0$ and $\hat{g}(0) > 0$ gives a smaller bound. The key step is a coercivity estimate for the Weil form, obtained by writing it as a truncated Wiener–Hopf form with a logarithmically growing symbol.

Introduction

Under GRH, the explicit formula bounds the analytic rank r_{an} of E by $W_E(g)/\hat{g}(0)$ for every admissible test function g; Mestre [7] introduced the method and Bober [3] made it practical with a fixed Fejér-type kernel. In [6] the test function is optimised for each curve within the span of $P_{2j}(x/L)$, $j < K$; the resulting *radix bound* $B_K(L) = 1/(v^{\top} G^{-1} v)$, where G is the Gram matrix of the Weil quadratic form, saturates numerically already for $K \approx 10$. Assuming GRH, this note proves that $B_K(L)$ converges to the best bound obtainable from continuous test functions supported in $[-2L, 2L]$ with non-negative Fourier transform, and identifies the limit and the extremal function, the *limiting radix kernel* $k_{E,L}$ (Theorem 5).

Throughout, $E/\mathbb{Q}$ is an elliptic curve of conductor N and we assume GRH for $L(E, s)$, normalised with centre $s = \frac{1}{2}$; its zeros are $\frac{1}{2} + i\gamma$. For $p \nmid N$ write $a_p/\sqrt{p} = \alpha_p + \beta_p$ with $\alpha_p \beta_p = 1$, $|\alpha_p| = 1$, and put $c_{p^k} = \alpha_p^k + \beta_p^k$; for $p \mid N$ put $c_{p^k} = (a_p/\sqrt{p})^k$. Let $w_n = \Lambda(n) c_n/\sqrt{n}$, $\hat{f}(t) = \int f(x) e^{itx}\, dx$, $f^{\star}(x) = \overline{f(-x)}$.

1. The Weil form as a Toeplitz form

For $f \in L^2(\mathbb{R})$ with support in $[-L, L]$ and $g = f * f^{\star}$ (so $\hat{g} = |\hat{f}|^2$, $\operatorname{supp} g \subset [-2L, 2L]$), the right-hand side of the explicit formula [6, Sect. 2] is

$$W_E(g) = g(0)\big(\log N - 2\log 2\pi\big) + \mathrm{Arch}(g) - 2 \sum_{n < e^{2L}} w_n \operatorname{Re} g(\log n),$$

$$\mathrm{Arch}(g) = 2\Big[-\gamma_{\mathrm{E}}\, g(0) + \int_0^{2L} \frac{g(0) - \operatorname{Re} g(u)}{e^u - 1}\, du - g(0) \log\big(1 - e^{-2L}\big)\Big],$$

with values in $(-\infty, +\infty]$. The same formula defines $W_E(g)$ for every continuous g with $\operatorname{supp} g \subset [-2L, 2L]$ and $\hat{g} \geq 0$; then $g(0) \geq |g(u)|$, so the integrand in $\mathrm{Arch}(g)$ is non-negative. For smooth g the explicit formula gives $W_E(g) = \sum_{\gamma} \hat{g}(\gamma)$ [7, 3].

Lemma 1. *For every such f,*

$$W_E(g) = \frac{1}{2\pi} \int_{\mathbb{R}} |\hat{f}(t)|^2 M(t)\, dt,$$

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$$M(t) = \log N - 2\log 2\pi + 2\operatorname{Re}\psi(1+it) - 2\sum_{n<e^{2L}} w_n \cos(t\log n),$$

where $\psi = \Gamma'/\Gamma$. M is real and even, and $|M(t) - 2\log(e+|t|)| \le C_{E,L}$ for all $t \in \mathbb{R}$.

Proof. From $\psi(z) = -\gamma_{\mathrm{E}} + \int_0^\infty \frac{e^{-u}-e^{-zu}}{1-e^{-u}}\,du$ we get $\operatorname{Re}\psi(1+it) = -\gamma_{\mathrm{E}} + \int_0^\infty \frac{1-\cos tu}{e^u-1}\,du$. Since $g(u) = \frac{1}{2\pi}\int |\hat f|^2 e^{-itu}dt$, we have $g(0) - \operatorname{Re} g(u) = \frac{1}{2\pi}\int |\hat f|^2(1-\cos tu)\,dt$, and $g(u) = 0$ for $u > 2L$; Tonelli (the integrand is non-negative) gives $\frac{1}{2\pi}\int |\hat f|^2\, 2\operatorname{Re}\psi(1+it)\,dt = \operatorname{Arch}(g)$, including the case where both sides are $+\infty$. The prime sum is finite. The bound follows from $\operatorname{Re}\psi(1+it) = \log|t| + O(t^{-2})$ for $|t| \ge 1$. □

So W_E is a truncated Wiener–Hopf (continuous Toeplitz) form with symbol M on the Paley–Wiener space. The symbol is not non-negative, but under GRH the form is. Let $\mu(t) = 2\log(e+|t|)$ and let $\mathcal{H}_L$ be the space of $f \in L^2(\mathbb{R})$ with support in $[-L, L]$ and

$$\|f\|_\mu^2 = \frac{1}{2\pi}\int |\hat f|^2\mu < \infty, \qquad Q(f,h) = \frac{1}{2\pi}\int \hat f\,\overline{\hat h}\,M, \quad Q(f) = Q(f,f),$$

and, once Q is known to be positive definite, $\|f\|_Q = Q(f)^{1/2}$. $\mathcal{H}_L$ is a Hilbert space, $|Q(f)| \le C\|f\|_\mu^2$, and $W_E(f*f^\star) = +\infty$ for $f \notin \mathcal{H}_L$. The indicator $\mathbf{1}_{[-L,L]}$ (which generates Bober's Fejér kernel $g = \mathbf{1}_{[-L,L]} * \mathbf{1}_{[-L,L]}$) and every polynomial restricted to $[-L, L]$ lie in $\mathcal{H}_L$.

Lemma 2. *The embedding $\mathcal{H}_L \hookrightarrow L^2(\mathbb{R})$ is compact.*

Proof. For $\|f\|_\mu \le 1$ the functions $\hat f$ are uniformly bounded and uniformly Lipschitz ($|\hat f'| \le \int |xf| \le L\sqrt{L}$), so a subsequence converges locally uniformly (Arzelà–Ascoli), while $\frac{1}{2\pi}\int_{|t|>T} |\hat f|^2 \le 1/(2\log(e+T))$ uniformly; by Plancherel the subsequence is Cauchy in L^2. □

Lemma 3. *For $f \in \mathcal{H}_L$, $\sum_\gamma |\hat f(\gamma)|^2 \le Q(f)$ (zeros with multiplicity); in particular $r_{\mathrm{an}}|\hat f(0)|^2 \le Q(f)$. If $Q(f) = 0$ then $f = 0$.*

Proof. For smooth f this is the explicit formula (with equality) and Lemma 1. For general $f \in \mathcal{H}_L$, dilate ($f(x/(1-\delta))$) and mollify; the approximants are smooth with support in $[-L, L]$ and converge to f in $\mathcal{H}_L$ (on the Fourier side the dilations are uniformly bounded on $L^2(\mathbb{R}, \mu\,dt)$ since $\mu(s/(1-\delta)) \le C\mu(s)$, and converge strongly on $C_c(\mathbb{R})$, which is dense there), hence Q converges, and $\hat f$ converges uniformly on $\mathbb{R}$. Fatou's lemma gives the inequality.

If $Q(f) = 0$, then $\hat f(\gamma) = 0$ for every ordinate. Under GRH the multiplicity of a zero at $\frac12 + i\gamma$ is $\ll \log(N(|\gamma|+3))/\log\log(N(|\gamma|+3))$: apply [4, Thm. 5] to the twist $L(s+i\gamma, E)$, which satisfies GRH, the Ramanujan bound and a functional equation, with analytic conductor $\asymp N(|\gamma|+3)^2$. Since the number of ordinates in $[-T, T]$ counted with multiplicity is $\frac{T}{\pi}\log\frac{NT^2}{(2\pi e)^2} + O(\log NT)$, there are $\gg T\log\log T$ distinct ones. An entire function of exponential type has $O(R)$ zeros in $|z| \le R$ (Jensen), so $\hat f \equiv 0$. □

Theorem 4 (coercivity)**.** *There is $\lambda_{E,L} > 0$ with $Q(f) \ge \lambda_{E,L}\|f\|_\mu^2$ for all $f \in \mathcal{H}_L$.*

Proof. Write $M = \mu + R$ with R bounded. Then $Q(f) = \langle (I+T)f, f\rangle_\mu$, where $\langle Tf, h\rangle_\mu = \frac{1}{2\pi}\int \hat f\,\overline{\hat h}\,R$; T factors through the compact embedding of Lemma 2, so it is compact and self-adjoint. By Riesz–Schauder [8, Thm. VI.15], $\sigma(I+T) \cap (-\infty, \frac12)$ consists of finitely many eigenvalues, each positive by Lemma 3. Hence $\min\sigma(I+T) > 0$. □

2. The limiting radix kernel is optimal

Theorem 5. *Assume GRH for $L(E,s)$ and let $L>0$.*

(i) *There is a unique $k=k_{E,L}\in\mathcal{H}_L$ with $Q(f,k)=\hat f(0)$ for all $f\in\mathcal{H}_L$; it is real and even. Writing $\mathcal{R}_{E,L}(w,z)$ for the reproducing kernel of the space $\{\hat f : f\in\mathcal{H}_L\}$ with norm $\|\cdot\|_Q$, we have $\hat k(z)=\mathcal{R}_{E,L}(z,0)$ and $\mathcal{R}_{E,L}(0,0)=\hat k(0)=Q(k)>0$.*

(ii) $B^*(L):=\inf\limits_{0\neq f\in\mathcal{H}_L}\dfrac{Q(f)}{|\hat f(0)|^2}=\dfrac{1}{\mathcal{R}_{E,L}(0,0)}$, *attained exactly by the multiples of k; in particular $r_{\mathrm{an}}\le\lfloor B^*(L)\rfloor$.*

(iii) *For every continuous g with $\operatorname{supp} g\subset[-2L,2L]$, $\hat g\ge 0$ on $\mathbb{R}$ and $\hat g(0)>0$,*

$$\frac{W_E(g)}{\hat g(0)}\ \ge\ B^*(L).$$

Thus among all explicit-formula bounds whose test function lies in this class, so that only the coefficients c_n with $n<e^{2L}$ enter, $B^*(L)$ is the smallest, and the radix bounds $B_K(L)$ converge to it by (iv).

(iv) *Let $B_K(L)=1/(v^{\top}G^{-1}v)$ be the bound computed in the basis $P_{2j}(x/L)$, $j<K$ [6, Sect. 3]. Then $B_K(L)$ is non-increasing in K, $B_K(L)\to B^*(L)$, and*

$$B_K(L)-B^*(L)=\frac{\|k-P_Kk\|_Q^2}{\mathcal{R}_{E,L}(0,0)\,\|P_Kk\|_Q^2},$$

where P_K is the Q-orthogonal projection onto $V_K=\operatorname{span}\{P_{2j}(x/L):j<K\}$.

Proof. (i) By Theorem 4, $(\mathcal{H}_L,Q)$ is a Hilbert space, and $\ell(f)=\hat f(0)=\langle f,\mathbf{1}_{[-L,L]}\rangle$ is continuous, with $\langle\cdot,\cdot\rangle$ the inner product of $L^2(\mathbb{R})$ ($|\ell(f)|\le\sqrt{L}\|f\|_\mu$). Riesz representation gives k. Since M is even and real, $k(-x)$ and $\overline{k}$ represent ℓ as well, so k is real and even. $Q(k)=\ell(k)$, and $Q(k)>0$ as $\ell\neq 0$.

(ii) Cauchy–Schwarz in $(\mathcal{H}_L,Q)$: $|\hat f(0)|^2=|Q(f,k)|^2\le Q(f)Q(k)$, with equality iff $f\in\mathbb{C}k$. The bound on r_{an} is Lemma 3.

(iii) As g is continuous at 0 and $\hat g\ge 0$, $\hat g\in L^1(\mathbb{R})$ (Fejér summation at 0). $\hat g$ is entire of exponential type $\le 2L$, bounded and non-negative on $\mathbb{R}$. Hence $\int\log^+\hat g(t)\,(1+t^2)^{-1}dt<\infty$, and by Akhiezer's theorem [1, 2], $\hat g=|F|^2$ with F entire of exponential type $\le L$; $F\in L^2(\mathbb{R})$, so $F=\hat f$ with $\operatorname{supp} f\subset[-L,L]$ (Paley–Wiener), and $g=f*f^{\star}$. If $f\notin\mathcal{H}_L$, then $W_E(g)=+\infty$. Otherwise $W_E(g)=Q(f)$ by Lemma 1, $\hat g(0)=|\hat f(0)|^2$, and (ii) applies.

(iv) G is the Gram matrix of Q on V_K (invertible since Q is positive definite) and $v_j=\ell(P_{2j}(\cdot/L))$, so $v=(2L,0,\dots,0)$ by orthogonality of the Legendre polynomials. On V_K the functional ℓ is represented by P_Kk, so $B_K=1/\|P_Kk\|_Q^2$ as in (ii); Pythagoras gives the identity, and B_K is non-increasing because the V_K are nested. Even polynomials are dense in $\mathcal{H}_L\cap\{f\text{ even}\}$: for $h\in C^1[-L,L]$, integration by parts gives $|\widehat{h\mathbf{1}_{[-L,L]}}(t)|\ll\|h\|_{C^1}/|t|$, hence $\|h\mathbf{1}_{[-L,L]}\|_\mu\ll\|h\|_{C^1}$; smooth even functions supported in $(-L,L)$ are dense, and are C^1-limits of even polynomials. With the norm equivalence of Theorem 4, $\|k-P_Kk\|_Q\to 0$. □

Remark. Part (iii) concerns certificates that use $\hat g\ge 0$ on all of $\mathbb{R}$, which is what one needs when the zeros are unknown. Certificates using further information (for example the parity or integrality of r_{an}, or signs of individual prime terms) are not covered. In [6], on the curve #4239 of the ICARM Elliptic Curve Rank Leaderboard [5] (rank ≥ 32), increasing K from 10 to 20 changes the bound only in the fifth decimal; by (iv) this is consistent with $\|k-P_Kk\|_Q$ being small for $K\ge 10$. A quantitative rate in (iv) would follow from regularity of k, which we do not pursue here.

Tools. PARI/GP, Arb and Rust (the radix-rank code of [6]) for the numerical values quoted from [6]; Claude (Anthropic). The author takes full responsibility for the content; all numerical results are reproducible from the code and certificates of [6].

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